

Development of Educational Game for Indonesian History Lesson Focusing on Indonesia Pre-Literacy Relics "Prehistoria" for Grade X High School Student

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Abstract—The development of game technology has opened up new possibilities in learning methods through educational games. However, in Indonesia, the application of this concept is lacking. One lesson that can benefit from educational games is Indonesian History in class X, with a focus on history in the pre-literacy period. This study aims to create a puzzle-adventure type of digital game that can help educate students about the pre-literacy period, developed with Unity, and uses WebGL as a target device while referring to Indonesia's revised 2013 national curriculum. The result of this research is the creation of a digital game that can be played on average of 254.36 seconds, with potential for educational use. A survey result shows that many of the ancient objects and some of the ages are well-recognized. However, the survey also shows many of the gameplay aspects, such as UI, gameplay features, and adventure elements, need improvements.

Index Terms—Game, History, Pre-literacy, Indonesia, Educational Games

I. INTRODUCTION

Advances in video game technology have led to the discovery of the use of video games beyond its original purpose as an entertainment medium. It introduces a new terminology called educational game used to improve learning materials in the world of education. Educational games are serious games that are used in education, while serious games themselves mean games designed, developed, and used outside of entertainment purposes [1]. Many kinds of research show that educational games become one of the great potentials for the future of education. However, most studies use curricula used in their respective research countries. One of the less-discussed subjects is Indonesian History, which is taught at the high school level as a compulsory subject, making it a viable choice for gamification. The research will cover game design, game creation, field testing, and findings throughout the study. Finally, this research will show what

future work potential for gamification for Indonesian History, especially in the pre-literacy period, in Indonesia. The reason the pre-literacy period selected is due to the period being the first period taught in the Indonesian History lesson in tenth grade. In turn, the main target would be tenth-grade students. This gamification act as a medium of education, along with existing methods.

II. RELATED WORKS

Serious games have been in research for about a decade. Therefore, there is a lot of research on the use of video game technology in education, starting from theoretic [2], designing educational games [3], and even to review the use of concepts in its applications [4] such as math [5], and English [6]. There is research in the field of Virtual Reality [7] [8] for the Indonesian History topic, but no one has done it in educational games. This study aims to fill the role of teaching Indonesian History with video game technology. The most similar study is the historical gamification of Malaysian History [9].

With comparison to [9], there are differences between the referenced research and this paper. First, the target device used is different, with [9] using the mobile application while this research uses the WebGL. Second, [9] refers to previously made games as a reference, while this research uses the game genre as reference for the game. Third, [9] does not elaborate on the game development process but compares them with existing games, while this research elaborates upon the development process.

III. METHOD

Figure 1 shows that the study began with the creation of a gameplay mechanic and followed by a game story to create a framework for the game, continued with the process of synergizing with the curriculum to select contents included

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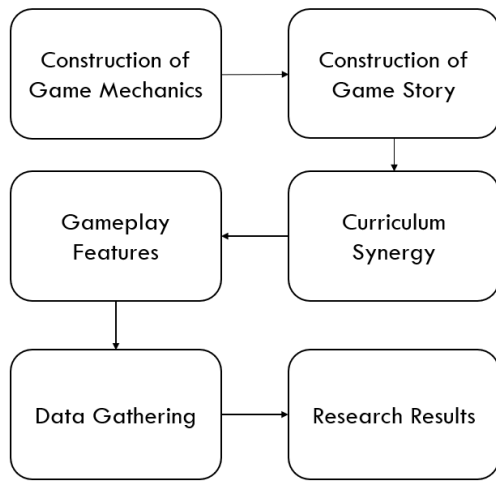


Fig. 1. Research Method

into the game. Next, the creation of Game Features during game development. Once the game is ready, Field Survey to collect player data. Finally, the data is analyzed and then summed into the Research Result, which is part of Result and Discussion.

A. Gameplay Mechanic

There are several mechanics in the game:

- 1) The game starts with all levels unlocked (for ease of testing)
- 2) Player and Enemies starts a predetermined location
- 3) Player can move left, right, up, and down
- 4) Player need to take a photo of objects at predetermined spots
- 5) Player must avoid Enemies
- 6) Player takes damage when hit by Enemies
- 7) When the player dies, the level restarts
- 8) When all objects' photo taken, the level continues to the next one

The general mechanic from the player perspective during one level in Figure 2. The game starts, then players check if they are at a target spot. If not in the target spot, they check their surroundings, and if there are enemies nearby, adjust for maneuver. When a player finds a target, they capture it, and capturing all of them clears the level. If a player dies, the level resets.

B. Gameplay Story

Become a time traveler who can explore various eras and learn firsthand about pre-literacy ages from items it left behind.

In this game, a player encounters several periods. Every period has challenges and puzzles that must be solved by the player. Here is a brief description of each level:

- 1) At the first level, players will explore the world in the early days of life, the Paleozoic era. Introduction level, so there are no obstacles other than the shape of the game map.
- 2) On the second level, players will explore the world of the Mesolithic era. At this level, players must also be able to avoid direct interaction with various ancient humans to maintain the time-space continuum.
- 3) On the third level, players will explore the world in the Neolithic era. At this level, players still need to

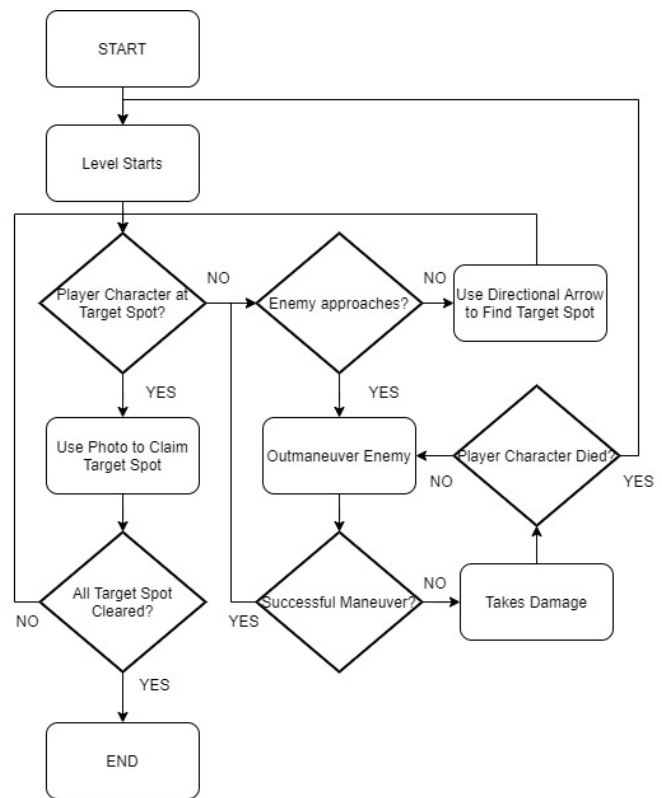


Fig. 2. Gameplay Flowchart

look for certain items and must avoid direct interaction with early humans. More ancient humans and complex challenges.

- 4) On the fourth level, players will explore the world in the Megalithic era. Still the same as before, but some items are harder to find, and also the challenges are increasingly complex. At this level also introduced new enemies who chase players in an attempt to stop the player from gathering photos of ancient objects.
- 5) At the last level, players will return to the present and need to bring the results of their time travel to the historical center for research. As the only non-preliteracy level, the only enemies are the chasing ones.

C. Curriculum Synergy

To adjust to the curriculum used in Indonesia, reference taken from "Erlangga" 10th Grade Indonesian History Book which is one of the books used for the Indonesian History lesson. According to the book itself, the pre-literacy period is the second chapter of the book, and the first chapter is the introduction of Indonesian History at the concept level.

The content selected from the book is as follows:

TABLE I
POTENTIAL OBJECTS ALONG WITH OBJECTS' AGE

Age	Objects
Paleozoic	Trilobites, Cephalaptis
Mesolithic	Hunting Period Tools, <i>Kjokkenmoddinger</i> , Cave Painting, <i>Abris sous roche</i>
Neolithic	Ax, Pickaxe, Planting
Megalithic	Homo Sapiens, Menhir, Ladder Punden, Grave Stone, Sarcophagus, Dolmen, Stone Statue, Kalamba, Waruga, Nekara, Moko

Now, as for the focus of the game, which is the Pre-literacy period. The pre-literacy period is a period where there is

no literacy. This period refers to a period where there's no writing has been discovered.

This period is further splits based on several "ages". The game itself focuses only on some of these "ages". The ages taken are Paleozoic where only a small amount of life exists. Then Mesolithic, that is also referred to as the "Hunting" age because people hunt animals for basic needs like food and clothing. Neolithic follows, referred to as the "Planting" or "Farming" age due to the existence of people planting vegetables and the likes. Afterward is the Megalithic, which is also referred to as the "Big Rock" age due to many objects and structures made from large rocks.

Objects in *game* are divided into several ages in the reference book, which reflects each age's alternative name. Potential objects can be seen in Table I. Of all these objects, they are sorted into Table II.

D. Game Features

Based on the Gameplay Flowchart in Figure 2, several features are needed to allow the game to function as expected.

1) *Health*: The feature is visible from Character Status in a long white color bar next to the portrait. The player's health will decrease when hitting enemies on the map. When the health decrease, a black colored bar will block portions of the white bar. This feature is tied to the "take damage" and "died" aspect in Gameplay Mechanic.

2) *Arrow Movement*: When a player pressed the button, the UI will run the character in the direction of the pressed arrow. Arrow Movement is tied to the movement in four directions mentioned in Gameplay Mechanic and is a key feature to play the game.

3) *Inventory*: Inventory starts empty at the beginning of each level when accessed by players. Every time a player gets an object at that level, Inventory will load said object. The numbers of objects in the Inventory will continue to increase along with the target object he gets.

4) *Photo*: It is a button that functions to activate the aspect of recording photos of the game. The code contained in the Character element controls the button. Figure 3 represents button's flowchart. Target Point communicates with this button's data.

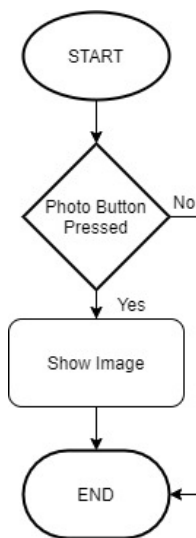


Fig. 3. Photo Button Flowchart

The target point is the point that marks the position that the player must reach. Each level has more than one target point,

except for the last level. The target point will display the image to the player when the player uses the photo capture button.

When taking a photo, a photo appears with additional information that shows the object name obtained by the player. The target works using the response to the Collider attached to the Target and Character objects.

TABLE II
LEVEL AND TARGET OBJECT NAMES

Level	Object Name
1	Trilobites
1	Cephalaptis
2	Hunting Period Tools
2	Kjokkenmoddinger
2	Cave Painting
3	Ax
3	Pickaxe
3	Planting
4	Homo Sapiens
4	Menhir
4	Nekara
4	Moko
5	Research Lab

Each level has one or more target points that must be taken by the player to complete the level. All objects are in table II. This Photo feature is tied with the "photo" aspect in Gameplay Mechanic and a key feature to finish the game.

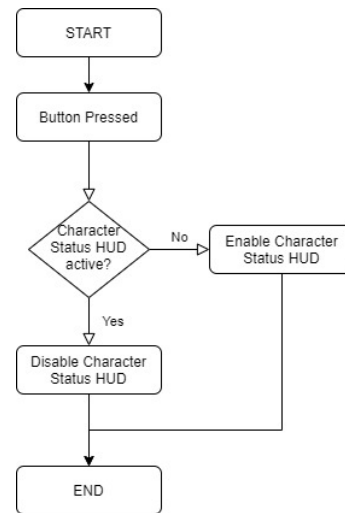


Fig. 4. Status Switch Button Flowchart

5) *Toggle Status*: It is a feature that works to determine whether Character Status will be displayed or not and serves as an extension feature to reduce distractions from the Health feature. It toggles the visibility of Character Status, as shown by the flowchart in Figure 4.

6) *Map*: Not player related, but a game feature. Each level has a different layout of the map that the player can traverse through. However, the map has several generalities that apply to all levels.

The map splits into three parts. The land map defines the playable area, while the water map restricts parts of the land map from being traversed. The fog map renders the map invisible when the player is not in range. Gameplay map is a combination of all three, like in Figure 5.

7) *Enemy*: Another game feature. This particular feature is to help implement the story aspect of the game. There

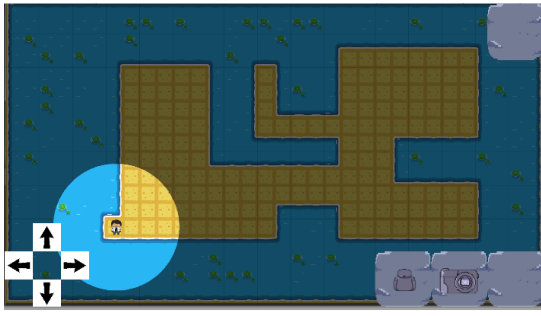


Fig. 5. Level One Final Map

are two kinds of enemies in the game. Each enemy type has different behaviors, which is represented by Figure 6.

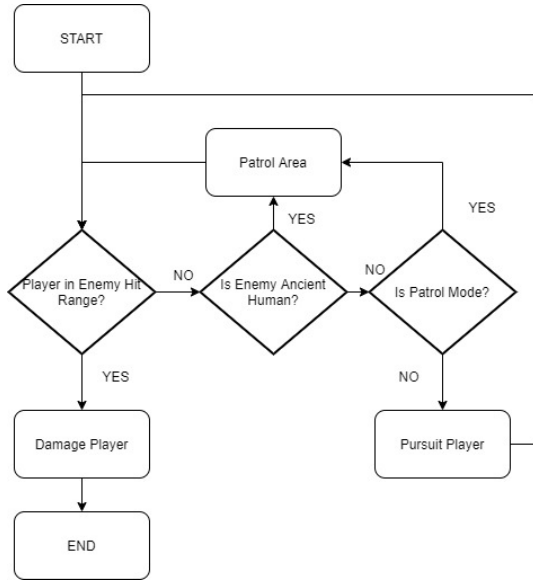


Fig. 6. Enemy Flowchart

The first enemy is the ancient humans. These kinds of enemies only patrol within their designated zones. The second enemy is the assassins. They hunt down players throughout the level. They will go into patrol mode periodically before continuing chase to the player.

E. Data Gathering

Via [10], user time data is taken. A survey to acquire additional data also executed. Users who take the survey are referred to as respondents, while users refer to all who played the game. There are a total of five respondents, 4 from 10th grade and 1 from 12th grade.

Below is the list of questionnaires:

- 1) Recognized Ancient Objects from Table II
- 2) Recognized Age from Table I
- 3) (1-5 Scale/Level) Level Difficulty
- 4) (Yes/No) Is Puzzle Element Recognized?
- 5) (Yes/No) Is Adventure Element Recognized?
- 6) (1-5 Scale) Is Loading Up speed fast enough?
- 7) (1-5 Scale) Is Game Core Mechanics easy to understand?
- 8) (Options in Table IX) What Features are recognized by the respondent during gameplay?
- 9) (1-5 Scale) Is Health Feature understandable?
- 10) (1-5 Scale) Is Arrow Movement Control easy to use?

- 11) (1-5 Scale) Is Inventory System Feature understandable?
- 12) (1-5 Scale) Is Photo Feature understandable?
- 13) (1-5 Scale) Is Toggle Status Feature understandable?
- 14) (1-5 Scale) Is UI Disrupts Gameplay?
- 15) (1-5 Scale) Does Map coloring represents Ages properly?
- 16) (1-5 Scale) Does the enemy provide an adequate challenge?
- 17) (1-5 Scale) Will a Tutorial help understanding the game better?

The goal of the first two questions is related to the understanding of the game as part of its' education role. The Level difficulty is to see the difficulty scaling of the game. For Puzzle and Adventure elements, to see if the respondent recognizes both elements. Loading speed, core game mechanics, UI disruption, enemy challenge, and tutorial are related to the gameplay experience of the player. Questions related to features are to see how recognizable the features from a player perspective. The map coloring is to help understand how well age and map correlates. The 1-5 scale used in this research refers to the 5-Point Likert Scale [11].

IV. RESULT AND DISCUSSION

The result divides into the gameplay time, educational aspect, and gameplay aspect. The gameplay time is from users playing the game, while both educational aspects and gameplay aspects are from questionnaires.

A. Gameplay Time

The gameplay time is to find out how long the game session is needed for students to finish the game on average.

$$m = \frac{\sum_k x}{k} \quad (1)$$

Formula 1 is an average formula where it takes all recorded time entries for the specific level and sums them all, denoted by $\sum_k x$, then divide them with the number of entries represented by k , giving the average of recorded time valued as m . This formula is to find out the time needed to play the game on average. The data itself acquired by having users play each level with their playtime recorded into the server. Then, the accumulated data of each level inserted into formula 1 to find the average for that level. The results in table III.

TABLE III
DATA AVERAGE PER LEVEL

Level	Play Time (Seconds)
1	38.96
2	63.2
3	60
4	64.1
5	28.1

B. Educational Aspect

Next is the educational aspect of the game. The following are data objects and times that were recognized from 5 respondents.

Of all the objects in the II table, not all respondents recognized all of the objects. Table IV shows how many respondents recognize each object.

Table V shows that some ages are not recognized by respondents. One respondent does not recognize any age.

TABLE IV
RECOGNIZED TARGETS

Level	Object Name	Respondent Recognized
1	Trilobites	1
1	Cephalaptis	1
2	Hunting Period Tools	3
2	Kjokkenmoddinger	3
2	Cave Painting	5
3	Ax	1
3	Pickaxe	2
3	Planting	4
4	Homo Sapiens	3
4	Menhir	4
4	Nekara	2
4	Moko	1

TABLE V
RECOGNIZED AGE

Age	Respondent Recognized
Paleozoic	1
Mesolithic	2
Neolithic	2
Megalithic	3
None	1

C. Gameplay Aspect

Now, focusing on the gameplay aspects. In the questionnaire, this begins from question 3 to 17. The result is as following:

TABLE VI
RESPONDENT OPINION ON LEVEL DIFFICULTY

Respondent	Difficulty for each Level
1	3-3-4-5-5
2	1-1-1-1-1
3	2-2-2-2-2
4	2-2-2-5-2
5	4-5-4-4-5

First, the distribution of difficulty levels uses a scale of 1 to 5, with 1 as Very Easy and 5 as Very Difficult. In general, each respondent provides one consistent number, with distributions as shown in Table VI. In general, respondents have varied perceptions of difficulty, with one respondent having a skewed difficulty for the 4th level, another respondent fluctuates slightly, one respondent having somewhat progressive, and two respondents having linear difficulty.

TABLE VII
RESPONDENT OPINION ON GENRE PRESENCE

Genre	Puzzle	Adventure
Respondents Agree	3	2
Respondents Disagree	2	3

Second, the elements of Puzzle and Adventure, with binary Yes or No. Respondents assumed that the Puzzle element was felt more than the Adventure element as shown by Table VII. 60 percent of respondents recognized the Puzzle genre, while 40 percent of respondents recognized the Adventure genre. This also shows that the Adventure element is not well-received by the respondents.

Third, respondents score the speed of loading up the game. There was one respondent who complained about the load speed that he deemed as very slow, while three respondents gave normal answers, and one gave a fast answer, shown by

TABLE VIII
RESPONDENT OPINION ON LOAD SPEED

Load Speed	Number of Respondents
Very Slow	1
Normal	3
Fast	1

Table VIII. On average, the loading speed is slightly below Normal, though it may vary from device to device used by the respondent.

Fourth, respondents' view of core mechanics. In general, respondents have difficulty understanding the mechanism of the game.

TABLE IX
RESPONDENT AWARENESS TO FEATURES

Feature	Respondent Recognized
Health	3
Arrow Movement	5
Inventory	4
Photo	5
Toggle Status	1

Next are the features related to the functional aspects game. From the features seen in Table IX, it can be seen that only 2 of the features that all respondents successfully identified, namely the Arrow Movement and the Photo feature. However, the Toggle Status is almost not recognized.

TABLE X
RESPONDENT OPINION ON EACH FEATURES

Understanding Difficulty	Health	Inventory	Photo	Toggle Status
Very Difficult	2	0	0	3
Difficult	2	2	0	1
Normal	1	2	0	0
Easy	0	0	4	1
Very Easy	0	1	1	0

Table X shows that the respondents having a difficult time understanding the Toggle Status. According to Table IX is the least aware, and the easiest to understand is the Photo feature. The health feature is quite hard to grasp despite being a core that is concerning and needs evaluation.

TABLE XI
RESPONDENT OPINION ON ARROW MOVEMENT EASE OF USE

Ease of Use	Number of Respondents
Very Difficult	3
Difficult	1
Normal	1

Do note that the Arrow Movement feature not inquired whether it is easy to understand because the focus of the feature is the ease of use, which according to them, is difficult to use, shown by Table XI.

TABLE XII
RESPONDENT OPINION ON UI DISRUPTION

UI Disruption	Number of Respondents
Very Disruptive	2
Disruptive	1
Neutral	2

When asked about the UI Disruption in the game. Two respondents strongly complained, one complained, and the

other two were neutral by the arrangement of game appearance as shown by Table XII. On average, the UI is quite disruptive. This means the UI can be improved to reduce disruptions during gameplay by the UI.

TABLE XIII
RESPONDENT OPINION ON MAP AND AGE RELEVANCE

Map and Age Relevance	Number of Respondents
Does Not Explain	2
Slightly Explain	1
Averagely Explain	1
Sufficiently Explain	1

When asked about the relevance of the map to the age period, two respondents gave answers that the map did not explain at all, one said the map slightly explains, one said the map moderately explains, and one said sufficiently explains as shown by Table XIII. This shows that the map does not indicate which age the player is playing.

TABLE XIV
RESPONDENT OPINION ON ENEMY ADEQUATE CHALLENGE

Enemy Challenge	Number of Respondents
Lacking	1
Moderate	1
Good	3

As for the Enemy, the respondent stated that they are challenging with three said quite challenging (Good), one moderately challenging (Moderate), and one not quite challenging (Lacking) as shown by Table XIV. This shows that the enemy provides above moderate challenge for the respondents, and does not need any alterations.

TABLE XV
RESPONDENT OPINION ON TUTORIAL NECESSITY

Tutorial Presence	Number of Respondents
Somewhat Useless	1
Somewhat Benefit	2
Greatly Benefit	2

Except for one respondent, they stated the tutorial feature would be useful in understanding how this game works. Two said it will greatly benefit while another two said it will somewhat benefit the game as shown by Table XV.

D. Discussion

Based on the data obtained, it finds that in general, the game-time needed to finish all levels estimated at 254.36 seconds. From the survey, only one object all respondents can identify. Respondent recognition of game age concerns, because one respondent does not recognize any of the ages. Adventure and Puzzle elements still need to be developed so they can be felt more by respondents. Currently, the Puzzle element is more recognized by the players, which means Adventure elements can use some improvements. A tutorial is needed that guides the player on how to play the game. The existence of a tutorial can help players recognize the features mentioned in Table IX. Several features, such as Enemies, Inventory, and Photos, does not need drastic changes because it is functioning and well recognized by the players. Others, however, need further addressing to make them more noticeable.

Respondents also complained about performance for game loading up. It might relate to development tool limitations and also the number of assets used. However, there should be some way to optimize the game further to reduce loading time.

V. CONCLUSION

From the discussion, there are several conclusions. First, the game time required by users to complete the entire game on average is 254.36 seconds (about 4 to 5 minutes). Some ancient objects and ages are difficult for respondents to recognize. Puzzle and Adventure elements still need to be developed further for recognition. There are still performance issues at the loading up phase of the game. Enemy, Photos, and Inventory Features are recognizable enough for the respondents, but other features need a tutorial to make it recognizable.

VI. LIMITATION AND FUTURE WORKS

The Indonesia Covid-19 pandemic prevents testing face-to-face controlled environment due to government restrictions and protocols at the time of research.

The current research is limited to Indonesian History. The tools used for game creation are Unity 2018, build on the WebGL platform deployed at [10], using 2D Art, with Erlangga's Class 10th Indonesian History Book used as a reference for game content.

There are many potential future works. The earliest future work is doing a full-pledged human-machine interaction and content testing of the game. In the game development aspects of this research, there are several potential improvements: more clarity on the objects, easy to understand features, more content within the realm of the pre-literacy period, and create a tutorial to improve the overall gameplay experience. Further, using a different reference book, different gameplay styles, and mechanics, and gamification of other topics within the realm of Indonesian History are also available options for future researches.

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